Solutions @ Tutorial 1

1

(a) $$z = 4x^2 - 8xy^4 + 7y^5 - 3$$

$$\frac{\partial z}{\partial x} = 8x - 8y^4$$

$$\frac{\partial z}{\partial y} = -32xy^3 + 35y^4$$

$$\frac{\partial^2 z}{\partial x^2} = 8 \quad ; \quad \frac{\partial^2 z}{\partial y^2} = -96xy^2 + 140y^3$$

$$\frac{\partial^2 z}{\partial x \partial y} = -32y^3$$

$$\frac{\partial^2 z}{\partial y \partial x} = -32y^3$$

(b) $$u = x^2 - 2xy + 2y^3$$

$$x = s^2 \ln t \quad ; \quad y = 2st^3$$

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial s}$$

$$= (2s^2 \ln t - 4st^3)(2s \ln t) + (-2s^2 \ln t + 24s^2t^6)(2t^3)$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial t}$$

$$= \left(2s^2 \ln t - 4st^3\right)\left(\frac{s^2}{t}\right)$$
$$+ \left(-2s^2 \ln t + 24 s^2 t^6\right)\left(6st^2\right)$$

(c) Say r be the radius
and h be the height

Then, volume $V = \pi r^2 h$

$$\frac{\partial V}{\partial r} = 2\pi r h \quad , \quad \frac{\partial V}{\partial h} = \pi r^2$$

$$\delta V \approx \frac{\partial V}{\partial r}\delta r + \frac{\partial V}{\partial h}\delta h$$

$$= 2\pi r h \ \delta r + \pi r^2 \delta h$$

Given, $r = 0.3\,m$; $h = 1\,m$

$\delta r = 1\text{ cm} = 0.01\,m$
$\delta h = 5\text{ cm} = 0.05\,m$

$$\delta V \approx 2\pi(0.3)(1)(0.01) + \pi(0.3)^2(0.05)$$
$$= 0.033\,m^3 \quad //$$

2.

$$V = \pi r^2 h$$

$$\frac{dV}{dt} = \frac{\partial V}{\partial r}\frac{dr}{dt} + \frac{\partial V}{\partial h}\frac{dh}{dt}$$

$$= 2\pi r h \frac{dr}{dt} + \pi r^2 \frac{dh}{dt}$$

Given, $\frac{dr}{dt} = 3 \text{ cm/yr}$; $\frac{dh}{dt} = 60 \text{ cm/yr}$

$r = 45 \text{ cm}$, $h = 600 \text{ cm}$

Thus, $$\frac{dV}{dt} = 2\pi(45)(600)(3) + \pi(45)^2(60)$$

$$= 8.9 \times 10^5 \text{ cm}^3/\text{yr}$$

3.

(a) At the top of the hill,

z is maximum

$$\frac{\partial z}{\partial x} = 0 \quad , \quad \frac{\partial z}{\partial y} = 0$$

$$\Rightarrow 2y - 6x - 18 = 0 \rightarrow (i) \quad \Rightarrow 2x - 8y + 28 = 0 \rightarrow (ii)$$

Solving (i) and (ii) yields:

$$x = -2 \, , \quad y = 3$$

Thus, the hill's height is

$$\max |z| = 72 \text{ m.}$$

(b)

$$z = 2xy - 3x^2 - 4y^2 - 18x + 28y + 12$$

$$\vec{\nabla} z = \frac{\partial z}{\partial x}\hat{x} + \frac{\partial z}{\partial y}\hat{y}$$

$$= (2y - 6x - 18)\hat{x} + (2x - 8y + 28)\hat{y}$$

$$\vec{\nabla} z \Big|_{x=1, y=1} = -22\hat{x} + 22\hat{y} = 22(\hat{y} - \hat{x})$$

$$\left| \vec{\nabla} z \Big|_{x=1, y=1} \right| = 22\sqrt{2} \quad \text{m/km}$$

Direction : Northwest

Q.4

The equation of motion is

$$m\frac{dv}{dt} = -mg - mkv^2$$

$$\Rightarrow \frac{dv}{g+kv^2} = -dt \rightarrow (1)$$

Say, the particle reach the heighest point after a time T

Integrating (1), we obtain

$$\int_u^0 \frac{dv}{g+kv^2} = -\int_0^T dt$$

$$\Rightarrow \frac{1}{\sqrt{gk}} \tan^{-1}\left(\frac{v\sqrt{k}}{\sqrt{g}}\right)\Bigg|_u^0 = -T$$

$$\Rightarrow T = \frac{1}{\sqrt{gk}} \tan^{-1}\left(\frac{u\sqrt{k}}{\sqrt{g}}\right) //$$

Q.5

The particle experiences a force $\vec{F} = qE_0\,\hat{j}$ when it is in the region $0 \le x \le b$.

The equation of motion is:

$$M\frac{d\vec{v}}{dt} = \vec{F} = qE_0\,\hat{j}$$

$$\vec{v} = v_x\,\hat{i} + v_y\,\hat{j}$$

Thus,
$$\frac{dv_x}{dt} = 0 , \qquad \frac{dv_y}{dt} = \frac{qE_0}{M}$$
$$\Rightarrow \boxed{v_x = A} \qquad \Rightarrow \boxed{v_y = \left(\frac{qE_0}{M}\right)t + B}$$

A and B are constants of integration.

Now, say the particle is at the origin at $t=0$. Then,
$$v_x = u , \quad v_y = 0 \quad \text{at} \quad t = 0$$
$$\Rightarrow \quad A = u \quad \text{and} \quad B = 0$$

So,
$$v_x = u , \quad v_y = \left(\frac{qE_0}{M}\right)t$$

Again, at $t=0$, $x=0$, $y=0$

Thus, from $\frac{dx}{dt} = u$ and $\frac{dy}{dt} = \left(\frac{qE_0}{M}\right)t$

we obtain: $\quad x = ut , \quad y = \left(\frac{qE_0}{2M}\right)t^2$

Eliminating t we obtain
$$y = \left(\frac{qE_0}{2Mu^2}\right)x^2$$

The angle through which the particle is deflected is the angle α shown in Fig. 1.
$$\tan\alpha = \left.\frac{dy}{dx}\right|_{x=b} = \frac{qE_0 b}{Mu^2}$$

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Q.6

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\frac{\partial r}{\partial x} = \frac{x}{r} \quad , \quad \frac{\partial r}{\partial y} = \frac{y}{r} \quad , \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\vec{\nabla} r = \frac{\hat{x}x + \hat{y}y + \hat{z}z}{r} = \frac{\vec{r}}{r} = \hat{r}$$

$$\frac{\partial r^n}{\partial x} = n r^{n-1} \frac{\partial r}{\partial x}$$

$$\frac{\partial r^n}{\partial y} = n r^{n-1} \frac{\partial r}{\partial y}$$

$$\frac{\partial r^n}{\partial z} = n r^{n-1} \frac{\partial r}{\partial z}$$

Thus,

$$\vec{\nabla} r^n = n r^{n-1} \vec{\nabla} r = n r^{n-1} \hat{r}$$

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Solution to practice problems

Q. 7

The equation of motion is :

$$m \frac{dv}{dt} = -\frac{GmM}{r^2}, \quad \begin{array}{l} M \equiv \text{Mass of Sun} \\ m = \text{mass of Earth} \\ r = \text{distance of Earth to Sun} \end{array}$$

Write: $\frac{dv}{dt} = \frac{dv}{dr}\frac{dr}{dt} = v\frac{dv}{dr}$

Thus,

$$v\frac{dv}{dr} = -\frac{GM}{r^2}$$

$$\Rightarrow \int v\, dv = -GM \int \frac{dr}{r^2}$$

$$\Rightarrow \frac{1}{2}v^2 = \frac{GM}{r} + A, \quad A \text{ is the integration constant}$$

Now when $r = R$, $v = 0$ (R is the radius of earth's orbit)

So, $A = -\frac{GM}{R}$

Thus, $$v^2 = 2GM\left(\frac{1}{r} - \frac{1}{R}\right)$$

$$\Rightarrow \left(\frac{dr}{dt}\right)^2 = 2GM\left(\frac{1}{r} - \frac{1}{R}\right) = 2GM\,\frac{R-r}{rR}$$

$$\Rightarrow \frac{dr}{dt} = \pm\sqrt{\frac{2GM}{R}}\sqrt{\frac{R-r}{r}}$$

Note that here $\frac{dr}{dt} < 0$ (as time increases the distance decreases)

So, $$\frac{dr}{dt} = -\sqrt{\frac{2GM}{R}}\sqrt{\frac{R-r}{r}}$$

Say T is the time taken by Earth to reach the Sun.

Then, $$\int_R^0 \sqrt{\frac{r}{R-r}}\,dr = -\sqrt{\frac{2GM}{R}}\int_0^T dt$$

(Substitute $r = R\sin^2\theta$, $dr = 2R\sin\theta\cos\theta\,d\theta$)

$$T = \sqrt{\frac{R}{2MG}}\int_0^{\pi/2}\left(\frac{\sin^2\theta}{1-\sin^2\theta}\right)^{1/2}(2R\sin\theta\cos\theta)\,d\theta$$

$$\boxed{T = \frac{1}{2}\pi\left(\frac{R^3}{2GM}\right)^{1/2}}$$

If you put the values, you should get $T \simeq 65$ days!

8. The equation of motion is

$$M \frac{dv}{dt} = - a e^{bv}$$

$$\Rightarrow \quad \frac{dv}{e^{bv}} = - \frac{a}{M} dt$$

Integrating we get:

$$- \frac{1}{b} e^{-bv} = - \frac{a}{M} t + c \quad \rightarrow (1)$$

At $t = 0$, $v = u$, so

$$c = - \frac{1}{b} e^{-bu} \quad \rightarrow (2)$$

Putting (2) in (1) we obtain:

$$\frac{1}{b} \left(e^{-bu} - e^{-bv} \right) = - \frac{a}{M} t$$

$$\Rightarrow \quad t = \frac{M}{ab} \left(e^{-bv} - e^{-bu} \right)$$

When the boat stops, $v = 0$. So, the corresponding time is

$$t = \frac{M}{ab} \left(1 - e^{-bu} \right) \quad //$$

9.

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial r}$$

$$= \cos\theta \frac{\partial u}{\partial x} + \sin\theta \frac{\partial u}{\partial y}$$

$$\rightarrow (i)$$

Differentiating (i) w.r.t. r

$$\frac{\partial^2 u}{\partial r^2} = \cos\theta \frac{\partial}{\partial r}\left(\frac{\partial u}{\partial x}\right) + \sin\theta \frac{\partial}{\partial r}\left(\frac{\partial u}{\partial y}\right)$$

$$= \cos\theta \left(\frac{\partial x}{\partial r} \frac{\partial^2 u}{\partial x^2} + \frac{\partial y}{\partial r} \frac{\partial^2 u}{\partial y \partial x} \right)$$

$$+ \sin\theta \left(\frac{\partial x}{\partial r} \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial y}{\partial r} \frac{\partial^2 u}{\partial y^2} \right)$$

$$= \cos^2\theta \frac{\partial^2 u}{\partial x^2} + \sin\theta \cos\theta \frac{\partial^2 u}{\partial x \partial y}$$

$$+ \sin\theta \cos\theta \frac{\partial^2 u}{\partial x \partial y} + \sin^2\theta \frac{\partial^2 u}{\partial y^2}$$

Also, $$\frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial \theta}$$

$$= -r \sin\theta \frac{\partial u}{\partial x} + r \cos\theta \frac{\partial u}{\partial y}$$

$$\frac{\partial^2 u}{\partial \theta^2} = r^2 \sin^2\theta \frac{\partial^2 u}{\partial x^2} + r^2 \cos^2\theta \frac{\partial^2 u}{\partial y^2}$$

$$- 2r^2 \sin\theta \cos\theta \frac{\partial^2 u}{\partial x \partial y}$$

$$- r \cos\theta \frac{\partial u}{\partial x} - r \sin\theta \frac{\partial u}{\partial y}$$

In this way you can show:

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

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